PURCHASE ORDER

IFIC-CSIC → ENGIONIC

PO-FD-VD-20260305-001

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Issuing Body: **IFIC-CSIC**
Supplier: **ENGIONIC**

1. Operating Conditions

Parameter	Value / Condition	Notes
Operating Temperature	86.5 – 293 K	Liquid Argon cryostat (DUNE FD-VD)
Applied Strain	Hanging mass ~200g at lead-out	Constant tension throughout operation
Electric Field	500 V/cm	Lead-in is outside the high-field region
Operational Lifetime	> 20 years	Continuous operation
Interrogation Window	1529 – 1568.2 nm	FBGs must remain within 1534.5–1566.5 nm at all times

2. Item 1 — FBG Fibre Sensors

2.1.1 Configuration 1: Physical Specifications

Parameter	Specification	Notes
Number of Fibres	7 fibres	
Fibre Type	SM1330-E9/125PI (Single Mode, Polyimide coating)	125 µm cladding, polyimide inner coat
Outer Coating	PEEK, 400 µm (± 10 µm) total OD	Cryogenic-compatible
Connector	FC/APC at lead-in	
Lead-in Length	269.0 mm (± 5 mm)	Origin: FC/APC ceramic ferrule
Lead-out Length	319.0 mm (± 5 mm)	Origin: fibre-end
Total Fibre Length	13686.5 mm (± 10 mm)	Origin: FC/APC ceramic ferrule; End: Cryostat bottom membrane + 100 mm slack
Number of FBG Sensors	28	Uniformly spaced in wavelength

2.1.2 Configuration 2: Physical Specifications

Parameter	Specification	Notes
Number of Fibres	10 fibres	
Fibre Type	SM1330-E9/125PI (Single Mode, Polyimide coating)	125 µm cladding, polyimide inner coat
Outer Coating	PEEK, 400 µm (± 10 µm) total OD	Cryogenic-compatible

Connector	FC/APC at lead-in	
Lead-in Length	244.0 mm (± 5 mm)	Origin: FC/APC ceramic ferrule
Lead-out Length	319.0 mm (± 5 mm)	Origin: fibre-end
Total Fibre Length	13661.1 mm (± 10 mm)	Origin: FC/APC ceramic ferrule; End: Cryostat bottom membrane + 100 mm slack
Number of FBG Sensors	28	Uniformly spaced in wavelength

2.2.1 Configuration 1: FBG Sensors Physical Design Positions (± 5 mm)

FBG Sensors Design Positions** (mm)											
S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**	**S8**	**S9**	**S10**	**S11**	**S12**
269	587	1087	1587	2087	2587	3087	3587	4087	4587	5087	5587
S13	**S14**	**S15**	**S16**	**S17**	**S18**	**S19**	**S20**	**S21**	**S22**	**S23**	**S24**
6087	6587	7087	7587	8087	8587	9087	9587	10087	10587	11087	11587
S25	**S26**	**S27**	**S28**								
12087	12587	13087	13368								

***All positions are referenced to FC/APC ceramic ferrule.*

2.2.2 Configuration 2: FBG Sensors Physical Design Positions (± 5 mm)

FBG Sensors Design Positions** (mm)											
S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**	**S8**	**S9**	**S10**	**S11**	**S12**
244	562	1062	1562	2062	2562	3062	3562	4062	4562	5062	5562
S13	**S14**	**S15**	**S16**	**S17**	**S18**	**S19**	**S20**	**S21**	**S22**	**S23**	**S24**
6062	6562	7062	7562	8062	8562	9062	9562	10062	10562	11062	11562
S25	**S26**	**S27**	**S28**								
12062	12562	13062	13343								

***All positions are referenced to FC/APC ceramic ferrule.*

2.3 Spectral Specifications – Common to Configuration 1 and Configuration 2

Parameter	Specification	Notes
Sensor Length	> 10 mm (homogeneous across sensors within ± 1 mm)	Required to achieve FWHM < 100 pm
Reflected Power	Optmised for interrogator model (between 20 % and 30%)	At interrogator input[1]
Insertion Loss	< 0.1 dB	Transmission outside Bragg peak
SLSR	> 20 dB	Main peak to highest sidelobe ratio
FWHM	< 100 pm	Ensures better than 30 fm resolution at 125 Hz / 1 min averaging (per polarisation)

[1] The reflectivity of the peaks will depend on the attenuation of the flanges and cables used in the readout chain, and the specifications in terms of delivered/received power of the interrogator. The interrogator models are the I4G and E4G (16 channels) from Optics11. Datasheets are available.

Birefringence	> 100 pm	Ensures proper peak separation between fast and slow components
Asymmetry Ratio [2](50%–10%)	< 3 %	Systematic centroid error minimisation
Sensor Spacing	1.15 nm	δλ = 31.05 nm / 28 sensors

2.4 Bragg Peak Design Positions – Common to Configuration 1 and Configuration 2 (± 50 pm)

Reference conditions: *20–22 °C | 45–65 % RH | Zero strain | Origin at FC/APC ceramic ferrule*

Bragg Peak Design Positions at Reference Conditions (nm)											
S1	**S2**	**S3**	**S4**	**S5**	**S6**	**S7**	**S8**	**S9**	**S10**	**S11**	**S12**
1535.00	1536.15	1537.30	1538.45	1539.60	1540.75	1541.90	1543.05	1544.20	1545.35	1546.50	1547.65
S13	**S14**	**S15**	**S16**	**S17**	**S18**	**S19**	**S20**	**S21**	**S22**	**S23**	**S24**
1548.80	1549.95	1551.10	1552.25	1553.40	1554.55	1555.70	1556.85	1558.00	1559.15	1560.30	1561.45
S25	**S26**	**S27**	**S28**								
1562.60	1563.75	1564.90	1566.05								

2.4 Summary of Manufacturing Tolerances

Parameter	Tolerance
Coating OD	10 µm
Lead-in Length	5 mm
Lead-out Length	5 mm
Total Length	10 mm
Physical Spacing	5 mm
Spectral Spacing	50 pm
FBG Length	1 mm

Notes

[1] *Lead-in and FBG1 are out of the high electric field region.*
[2] *Sensor position origin is defined at the FC/APC connector ceramic ferrule.*
[3] *Insertion loss refers to the transmission of the FBG sensor to the incident spectrum it is not designed to reflect.*
[4] *SLSR: ratio between the main reflection peak and its surrounding sidelobes.*
[5] *Bragg peak positions are defined at 20–22 °C, 45–65 % RH, atmospheric pressure and no transverse load.*
[6] *Interrogation window: 1529 – 1568.2 nm.*

[2] $A_{50-10} = \frac{|W_{L,50-10}-W_{R,50-10}|}{W_{L,50-10}+W_{R,50-10}}$ being $W_{L,50-10} = \lambda_{L,10} - \lambda_{R,50}$ and $W_{R,50-10} = \lambda_{R,50} - \lambda_{L,10}$ the widths of the spectrum between 50% and 10% of the maximum, respectively, at each side.